1. Supervised Model Training

1.1. Prior Probabilities of the Classes in the Dataset

```
P(0) NON MALICIOUS is 0.8
P(1) SCAM is 0.2
```

The prior probabilities for the class 0 and class 1 are 0.8 and 0.2 respectively.

1.2. Most Probable Words in Each Class

```
Top 10 words in class 0 and their probabilities:
```

word	count	prob
.	1416	0.0793
,	464	0.026
?	456	0.0256
u	337	0.0189
...	334	0.0188
!	306	0.0172
..	266	0.0149
;	234	0.0132
&	233	0.0131
go	198	0.0111

```
Top 10 words in class 1 and their probabilities:
```

word	count	prob
.	519	0.0565
!	223	0. 0244
,	215	0.0235
call	188	0.0205
£	127	0.0139
free	96	0.0105
∠	83	0.0091
2	80	0.0088
&	79	0.0087
?	77	0.0085

The tables above show that there is a significant overlap between the 10 most probable words in each class, such as ., !, ?, &, to name a few, which exist in both tables.

1.3. Words Most Strongly Predictive of Each Class

```
Top 10 most predictive words for NON-MALICIOUS (class 0):
;: 0.0165
...: 0.0174
gt: 0.0185
lt: 0.0187
:): 0.0209
ü: 0.0313
lor: 0.0347
ok: 0.0405
hope: 0.0405
d: 0.0474

Top 10 most predictive words for SCAM (class 1):
prize: 99.0284
tone: 64.0772
£: 49.7084
select: 46.6016
claim: 45.9543
paytm: 36.8929
code: 34.9512
award: 32.0386
won: 31.0677
18: 29.1260
```

These values are ranked by calculating a R value, defined as:

$$R = p_{c1,i}/p_{c2,i}$$

The motivation behind using these strongly predictive words is to improve model performance. It is especially useful in feature selection, to reduce noise and improve the efficiency of the model.

The results shown above are reasonable in the context of SMS-phishing. Scam words are often engineered to entice receivers to click on the link of malicious website unknowingly. The most predictive words associated with the scam classification are associated with winning some sort of prize, such as the ‘prize’, ‘select’, ‘award’, 'won’, etc.

It will be possible to distinguish the two classes using the Multinomial Naïve Bayes model, since the presence of distinctive, and high-R words allows the model to effectively learn the patterns in the data, to make accurate predictions. The results reflect class specific language patterns; for instance, informal texting patterns such as the smiley face, or colloquialisms like ‘lor’ are considered strongly predictive for the non-malicious class.

2. Supervised Model Evaluation

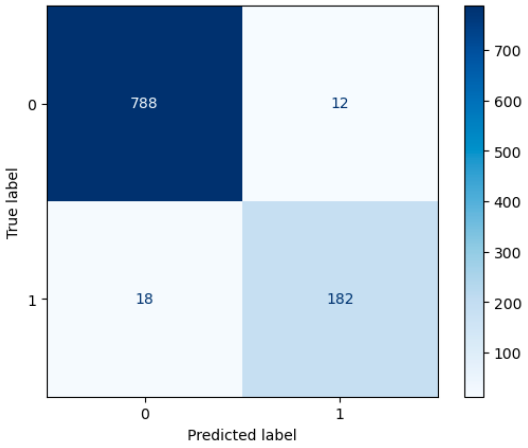
2.1. Reporting the Overall Accuracy of the Classifier

```
Baseline Model Accuracy: 0.975
              precision    recall  f1-score   support

           0       0.99      0.98      0.98       800
           1       0.93      0.95      0.94       200

    accuracy                           0.97      1000
   macro avg       0.96      0.97      0.96      1000
weighted avg       0.98      0.97      0.98      1000
```

Confusion Matrix:



This baseline model was computed by fitting the Multinomial Naïve Bayes model from the sklearn library, over the training dataset. This was then evaluated by using the test set provided. It is apparent that the model performs well, showing an accuracy of 0.975. The F1 score shows a strong balance between precision and recall, suggesting that this model is a reliable baseline. The precision in class 1 is slightly less than in class 0, demonstrating that class 1 is slightly more prone to false positives for class 1 compared to class 0. This could be since the training data available for class 1 is significantly less than class 0, as mentioned in section 1.

2.2. How Often the Model Encountered Out Of Vocabulary (OOV) Words in the Test Set.

```
Number of OOV words: 94
Number of test vocab words: 1641
Number of train vocab words: 2006

Number of test items containing ONLY OOV words: 0
No test items skipped
```

A test item is skipped when it contains no words from the learned vocabulary. Evidently, as there are no test items consisting only of OOV words, it can be deduced that there are no test items that are skipped entirely.

2.3. Instances Classified with High and Low Confidence.

Text	True Label	Predicted	R
Classified SCAM with high confidence			
. 4 + call £ - * holiday & …	scam	scam	1.3465287934116951e+20
. 3 4 + ! call : £ offer * holiday & urgent 18…	scam	scam	1.2795114910846806e+20
. . . , please order text call / :…	scam	scam	1.1410606356946185e+20
. call £ £ guarantee won customer …	scam	scam	9.044675823625771e+19
. . 2 free u + ! 1st / / wk …	scam	scam	7.535828315057827e+19
Classified NON-MALICIOUS with high confidence			
time : rs. transaction number & & …	non-malicious	non-malicious	0.0
? ? ? ? .. .. u u …	non-malicious	non-malicious	0.0
. every & & & & …	non-malicious	non-malicious	0.0
, get like second half & …	non-malicious	non-malicious	0.0
u , , lor ... ... …	non-malicious	non-malicious	0.0
Boundary cases (R ≈ 1)			
. call dear	non-malicious	non-malicious	0.98
. reply glad	non-malicious	non-malicious	0.96
? ur * just alrite sam	non-malicious	scam	1.08
. . tell return re order	non-malicious	scam	1.07

Non-malicious instances appear to be more easily distinguishable by the model. In most of the boundary cases where R ≈ 1; non-malicious cases are more correctly classified as class 1. This suggests that the model is more confident and consistent in detecting non-malicious messages. This observation aligns with the class distribution described in section 1, that non-malicious messages are more prevalent in the training set, which evidently causes the model to be better at learning these patterns.
This is also reflected in section 2.1, where the number of false negatives and false positives is slightly higher for class 0. This outcome is expected and highlights the importance of dataset representation in model performance.

3. Extending the Model with Semi-supervised Training: Label Propagation

Initially, the training dataset was split into a 80-20 ratio, where 80% of the dataset was used to train, and the remaining was used to validate the model.

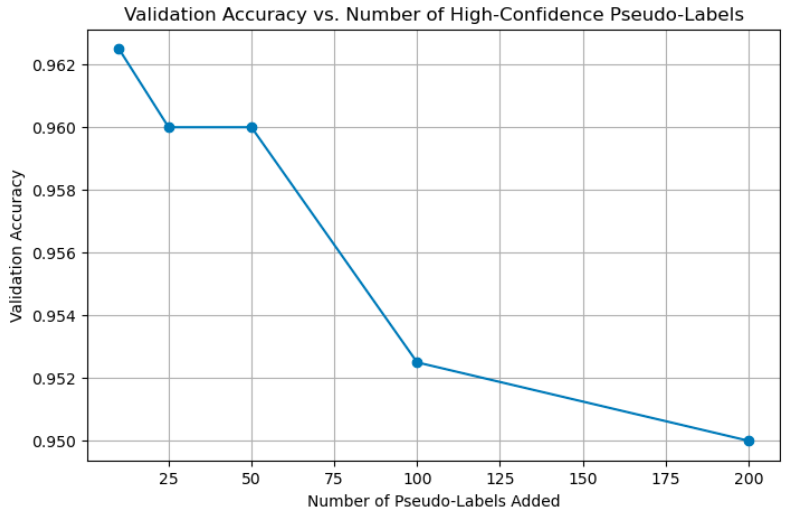
```
Q3-style Accuracy: 0.9775
              precision    recall  f1-score   support

           0       0.99      0.98      0.99       322
           1       0.92      0.97      0.94        78

    accuracy                           0.98       400
   macro avg       0.95      0.98      0.97       400
weighted avg       0.98      0.98      0.98       400
```

The figure above displays the accuracy and classification report of the base model with 'ground truth' labels. This was where the unlabeled dataset was given labels by having the initial baseline model predict its potential classes, which were then combined and added to the training set.

3.1. Confidence Labels



The model was then tested to see how it performed with only the 'n' most confident ground truth labels. A slight deterioration of validation accuracy can be observed from the graph, as the number of pseudo-labels added increases. This could be due to a variety of factors. For instance, although the labels added are those with the 'n' highest confidence, the model could still be confidently wrong. As more of this pseudo-labeled data is added, the chance of introducing noisy or mislabeled data increases. Additionally, the most confidently predicted pseudo-labels might only provide redundant information that the model has already learned. Many of the messages are composed of similarly structured text, or with the same words, which could add noise instead of being more useful to the model.

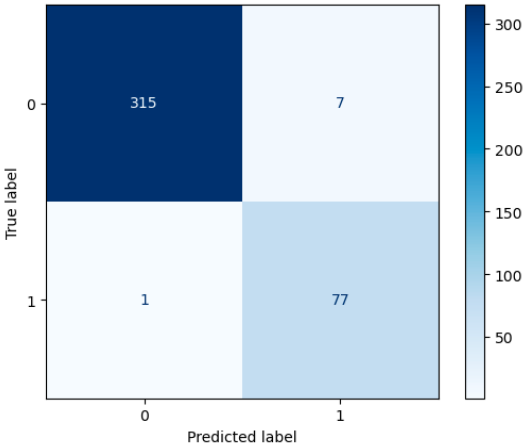
3.2. Iterative Model

An iterative method was implemented to extend the model with semi-supervised training.

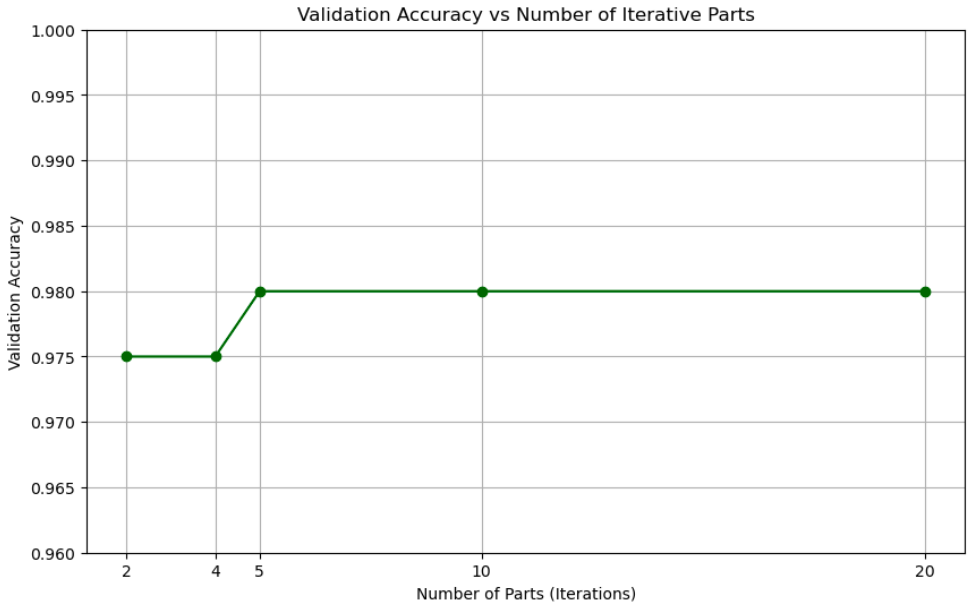
```
Final Accuracy after 5 iterations: 0.98
              precision    recall  f1-score   support

           0       1.00      0.98      0.99       322
           1       0.92      0.99      0.95        78

    accuracy                           0.98       400
   macro avg       0.96      0.98      0.97       400
weighted avg       0.98      0.98      0.98       400
```



This involved using the original model to predict the unlabeled data, splitting the unlabeled data into n parts, combining it with the training data, and retraining the model with this newly combined dataset. This retrained model was then used to train the next section, until the rest of the data had been trained.



The graph above suggests that the iterative model does provide varying results on its performance, but it plateaus after the data set is sectioned off into at least 5 parts. This could be because by the time the model has been retrained with pseudo-labels from the first few chunks (e.g., 5 parts), it has already seen the most useful and confident patterns in the unlabeled data. Furthermore, since the model is already performing well by 5 partitions, there is limited room for further improvement.

3.3. Findings

Results

Model	Accuracy
Baseline model	0.975
Base model with 'ground truth' labels	0.9775
Model with only high confidence labels	0.9625
Iterative model (split into 5)	0.98

The models shown in the table above were tested on the validation set.

The graph performs better when all the unlabeled data is used, compared to when only the high confidence labels are included. This suggests that even the lower-confidence samples can expose the model to more varied patterns that can provide more information to predict the classes correctly.

```
Top 1999 Confident Pseudo-Labels
Accuracy: 0.9575
              precision    recall  f1-score   support

           0       0.98      0.97      0.97       322
           1       0.89      0.90      0.89        78

    accuracy                           0.96       400
   macro avg       0.93      0.93      0.93       400
weighted avg       0.96      0.96      0.96       400
```

Here we can see that when all the pseudo-labels are added, the model improves itself slightly, supporting the fact that the model also benefits from volume of training data. Semi-supervised learning is most effective when the confidence is coupled with representativeness.

4. Semi-supervised Model Evaluation

```
Q3 on Test Set:
Accuracy: 0.976
              precision    recall  f1-score   support

           0       0.99      0.98      0.98       800
           1       0.92      0.96      0.94       200

    accuracy                           0.98      1000
   macro avg       0.96      0.97      0.96      1000
weighted avg       0.98      0.98      0.98      1000
```

This final Q3 model was based on the iterative model on 5 partitions, as it had the largest recall score for class 1, with a score of 0.99, as seen in section 3.2. This Q3 has a higher recall for class 1 compared to the original baseline model, which had a recall of 0.95. This is imperative in the context of detecting scams as it is safer to label non-malicious texts as scam, to minimize false negatives.

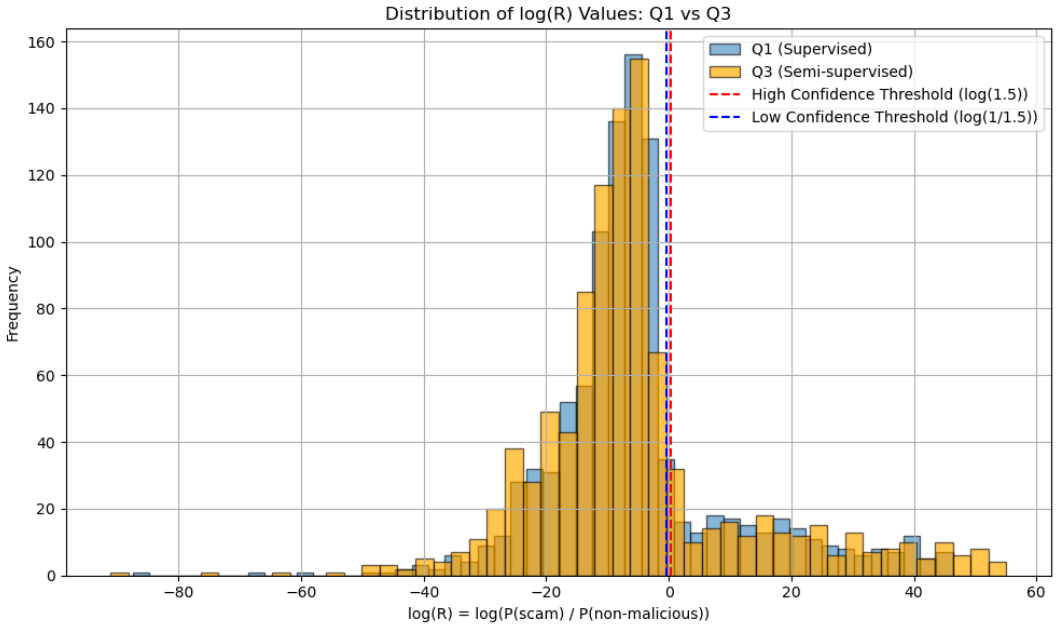
Text	True Label	Predicted	R
Classified SCAM with high confidence			
. . 2 free u + ! 1st / / wk…	scam	scam	8.650840289773109e+23
2 ur : award £ £…	scam	scam	8.481461299729145e+23
. call £ £ guarantee won…	scam	scam	1.916275941043867e+23
. . . , please order text call /…	scam	scam	1.3321113227055915e+23
. . . . . , weekend ! call…	scam	scam	.622714392878916e+22
Classified NON-MALICIOUS with high confidence			
time : rs. transaction number & & …	non-malicious	non-malicious	0.0
? ? ? ? .. .. u u …	non-malicious	non-malicious	0.0
. every & & & & …	non-malicious	non-malicious	0.0
, get like second half & …	non-malicious	non-malicious	0.0
? ?	non-malicious	non-malicious	0.0

The semi-supervised model has resulted in a subtle shift in the model's internal representation, which is reflected in both the model confidence changing from a 0.975 to 0.976, and the predictive features that it learns. For example, the R values went from ~1e+20 in the supervised baseline to ~1e+23 in the semi-supervised model. This demonstrates the model's increased confidence in its scam classifications. Similarly, messages classified as non-malicious still have very low R values, thus reinforcing the model's confidence in correctly identifying non-malicious messages.

```
Top 10 predictive words for NON-MALICIOUS in semi-supervised model:
lt: 0.0120
gt: 0.0122
;: 0.0157
ü: 0.0187
...: 0.0218
da: 0.0232
lor: 0.0235
ok: 0.0248
:): 0.0263
wat: 0.0338

Top 10 predictive words for SCAM in semi-supervised model:
prize: 215.6498
tone: 121.8022
claim: 88.5229
code: 73.8800
1000: 71.8833
150p: 71.8833
award: 70.8849
guaranteed: 69.8865
100: 59.9027
paytm: 57.9060
```

Furthermore, new highly predictive words have emerged, especially in class 1. Words like “1000”, “150p”, “guaranteed”, “100” now appear in the top predictive terms, which were not in the baseline model. This implies that the model is learning to strongly associate monetary and numeric terms with the scam class after being exposed to additional pseudo-labeled samples. For the class 0, the colloquial terms mentioned from section 1 are still prevalent and rank highly, but additional words such as “wat” and “da” have now gained predictive power. This reflects the model’s broader understanding of casual texting behaviors.



The graph above demonstrates the distribution of log(R) values of the baseline model (Q1) against the semi-supervised model (Q3). The distribution of both possesses a left-skewed shape, asserting that most texts are confidently non-malicious, which is expected due to the class distribution as outlined in section 1. It can also be seen that Q3 has a slight right tail, which suggests that it is more confident in detecting scams. Additionally, the region around $\log(R) \approx 0$ (between blue and red dashed lines) is sparser in Q3, meaning there is more clarity in decision boundary cases. The improved decision making of the Q3 model is further supported by how the histogram shows more spread to the left and right compared to the blue. Q3 is more confident in its predictions, outputting stronger probabilities for both classes.

The semi-supervised training has allowed the model to become more confident in its predictions. For both classes, it is learning new, more discriminative vocabulary for both classes, and better captures nuanced patterns in the data, such as informal or spam-like phrasing that were not as predictive in the supervised training set. Overall, these findings support the notion that semi-supervised learning, even with modest accuracy gains improve the model's ability overall to represent the data.